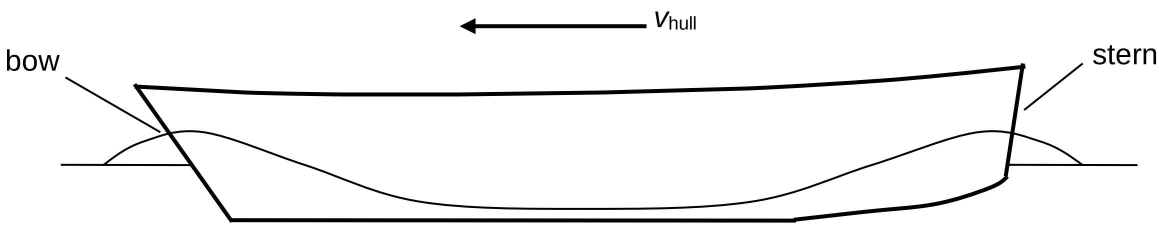
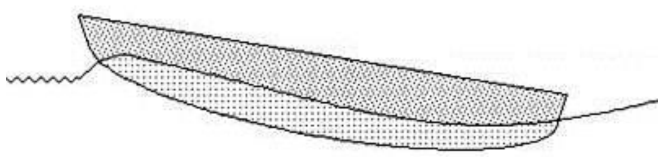
7(a)(i) Wavelength is defined as the distance between two successive crests or troughs of a progressive wave.

(ii)



(iii) Above hull speed, the wavelength of the bow wave increases such that the crests are no longer at the stern. The boat will be tilted, resulting in a larger drag.
OR
The driving force of the boat engine is also no longer horizontal, leading to inefficiency.



(b)(i) $v_{hull} = 4.35 \text{ m s}^{-1}$

$\lg v_{hull} = 0.638$

(ii)

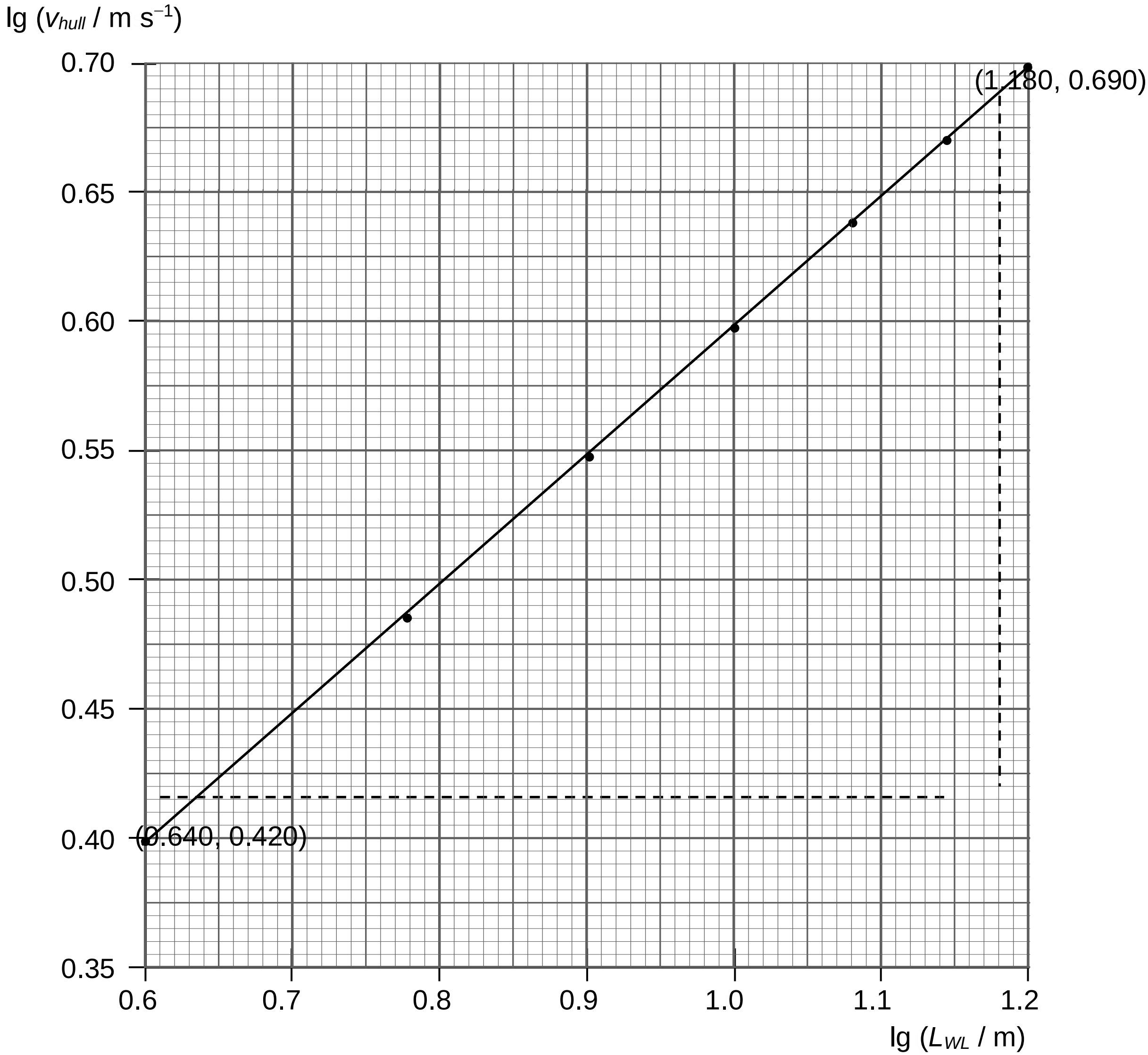


Fig. 6.3

(iii) n = gradient of best fit line

From graph, gradient = $\dfrac{0.690 - 0.420}{1.180 - 0.640} = 0.500$

Hence n = 0.500

(iv) $\lg v_{hull} = n \lg L_{WL} + \lg k$

Using (0.640, 0.420),

$0.420 = 0.500\ (0.640) + \lg k$

$\lg k = 0.100$

$k = 1.26$

(v) From $v_{hull} = k\ L_{WL}{}^{n}$ and n = 0.500

Base unit of k = $\dfrac{\text{m s}^{-1}}{\text{m}^{0.500}} = \text{m}^{0.500}\ \text{s}^{-1}$

(c)(i) From $v_{hull} = k\ L_{WL}{}^{n}$, n = 0.500 and k =1.26

$v_{hull} = k\ L_{WL}{}^{n} = 1.26 \times 110^{0.500} = 13.2\ \text{m s}^{-1}$

(ii) Warships are designed for combat operations. Having the highest possible maximum speed has priority over moving efficiently.